Superpixel-Level Target Discrimination for High-Resolution SAR Images in Complex Scenes

Zhaocheng Wang, Lan Du, *Senior Member, IEEE*, and Hongtao Su

***Abstract*—Traditional synthetic aperture radar (SAR) target discrimination methods are implemented at the chip-level, which may have good discrimination performance in simple scenes but may lose the effectiveness in complex scenes. To improve the discrimination performance in complex scenes, this paper proposes a superpixel-level target discrimination method directly in high-resolution SAR images. The proposed discrimination method mainly contains three stages. First, based on the superpixel-level target detection results, we describe each superpixel via the multilevel and multidomain feature descriptor, which can reflect the differences between targets and clutter comprehensively. Second, we employ the support vector machine as the discriminator to obtain the discriminated target superpixels. Finally, we cluster the discriminated target superpixels and extract the target chips from the original SAR image based on the clustering results. The experimental results based on the miniSAR real SAR data show that the proposed discrimination method has about 25% higher F1-score than the traditional discrimination methods.**

***Index Terms*—Bag-of-visual-words (BoVW), complex scenes, multilevel and multidomain (MLMD), superpixel, synthetic aperture radar (SAR), target discrimination.**

I. Introduction

With the rapid development of synthetic aperture radar (SAR) imaging technology, high-resolution SAR images have been widely employed in military and civilian applications. One of the important applications is the automatic target recognition (ATR), which includes three stages [1], namely, detection, discrimination, and classification. The detection stage selects the candidate target chips (small SAR images contain true targets or clutter false alarms) from the whole scene via the two-parameter constant false alarm rate (CFAR) [1] detection method. The discrimination stage removes the natural clutter chips from the candidate target chips to reduce the cost of the following classification stage. The classification stage rejects the man-made clutter chips and classifies the remaining chips by target type. There are numerous existing studies that focus on the detection stage [2]–[8] and the classification stage [9]–[14], but only a few of the studies are found that focus on the discrimination stage. However, many problems have still not been solved in the discrimination stage. In this paper, we focus on the discrimination stage, especially for complex scenes.

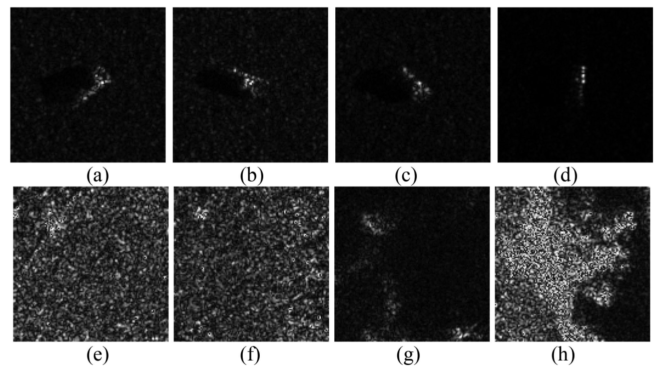

Fig. 1. Candidate target chips from the MSTAR dataset. (a)–(d) True target chips. (e)–(h) Clutter chips. Note that here there is only a single target at the center of each target chip, and the clutter in each clutter chip is rather simple.

Traditional SAR target discrimination methods are implemented at the chip-level [1], [15], i.e., they regard the candidate target chips that are extracted in the detection stage as the elementary units for target discrimination. Here, the classical discrimination features (CDFs), i.e., the old Lincoln features [16], new Lincoln features [17], Bhanu's features [18], and Gao's features [19], are usually extracted from each candidate target chip for SAR target discrimination. These CDFs are extracted based on the target segmentation results or the target boundary. The performance of the traditional SAR target discrimination methods, which are based on the candidate target chips and the CDFs, may be good in some simple scenes, such as the moving and stationary target acquisition and recognition (MSTAR) public release dataset (see Fig. 1). However, in complex scenes, such as the miniSAR data set (see Fig. 2), the traditional SAR target discrimination methods may have the following problems.

1) *Chip problem:* Since the chips have the same square shape and the same size, there may exist single target, multiple targets, and partial targets cases in a true target chip [see Fig. 2(a)–(d)], which results in large intraclass variations. In addition, the true target chips usually contain many backgrounds, which interfere with the feature extraction of the targets. Thus, it is difficult to have a good discrimination performance for SAR images in complex scenes using chips.

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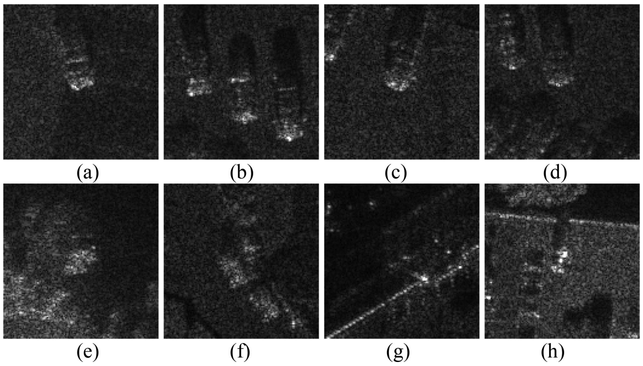

Fig. 2. Candidate target chips from the miniSAR dataset. (a)–(d) True target chips. (e)–(h) Clutter chips. Note that here (a) contains a single target, (b) contains multiple targets, (c) and (d) contain partial targets, and (e)–(h) contain some strong clutter, such as the building clutter.

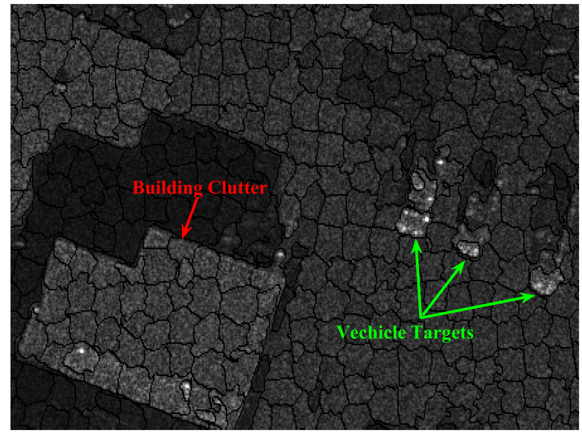

Fig. 3. Superpixel segmentation result of a small SAR image.

2) *Feature problem:* Some CDFs, which reflect target's size and shape, need segmentation preprocessing to obtain the target region. However, the segmented target region varies with the segmentation methods; thus, the corresponding features are not robust especially for complex scenes. Moreover, the CDFs are some relatively rough features that are usually used to distinguish the true targets from the natural clutter. They usually cannot distinguish the man-made clutter from the candidate target chips, due to their rough and noncomprehensive feature description.

To deal with complex scenes, Wang and Liu [20] recently proposed a new SAR target discrimination method based on the bag-of-visual-words (BoVW) model with dictionary learning. Here, the BoVW model mainly includes the local feature extraction stage and the local feature coding stage, and it can obtain some robust features without segmentation preprocessing. Although they have obtained some good discrimination results, they have the following issues. First, their proposed target discrimination methods are still at the chip level, which may lead to large intraclass variations in complex scenes as mentioned previously. Second, they employed SAR scale invariant feature transform (SAR-SIFT) [24] features in the local feature extraction stage, which only reflects the local characteristics of the objects in the texture domain. Third, they used dictionary learning and sparse coding in the local feature coding stage, which is very time-consuming.

According to the above analysis, in this paper, we propose a superpixel-level target discrimination method directly in high-resolution SAR images. The major contributions of our proposed discrimination method are summarized as follows.

1) *We use the superpixels rather than the chips as the elementary units for SAR target discrimination.* The superpixels are local coherent small regions obtained by over segmentation and their shapes can adhere well to object boundaries [21]. In the optical image field, some studies [22], [23] have applied the superpixels to target detection and classification. Recently, in the SAR image field, some studies [6], [7] used the superpixels as the elementary units for target detection, since these are more robust to the speckle noise than the pixels. Nevertheless, the superpixels have not been employed for SAR target discrimination. In this paper, we take the superpixels as the elementary units for SAR target discrimination. Moreover, to extract the discriminative features for each superpixel, we utilize the synthetic information from the current superpixel and its nearby multiple superpixels, rather than only utilizing the information from the single chip as done by the traditional discrimination methods. Fig. 3 shows the superpixel segmentation result of a small SAR image. As shown in Fig. 3, both the vehicle targets and the building clutter are segmented into several superpixels having a flexible shape, and each superpixel belongs to only one target or part of the background, not stretching over both of them, which is different from the chips shown in Fig. 2. Therefore, the superpixels are more suitable for SAR target discrimination than the chips in complex scenes.

2) *We propose the multilevel and multidomain (MLMD) feature descriptor for the superpixels based on the BoVW model.* Multilevel refers to the pixel-level, the superpixel-level, and the object-level. Here, the pixel-level features are obtained by some dense low-level descriptors; the superpixel-level features are obtained by the local feature coding based on k-means clustering and Euclidean distance calculation; the object-level features are obtained by combining the contextual information of the current superpixel in the high-resolution SAR image. Multidomain refers to that we extract the pixel-level features not only in the texture domain but also in the multiscale amplitude domain and the subaperture scattering domain. Therefore, the superpixels are described comprehensively via the proposed MLMD feature descriptor, which can facilitate to discriminate the targets from the clutter.

The remainder of this paper is organized as follows. Section II gives an overview of the proposed SAR target discrimination method. Section III explains the proposed MLMD feature descriptor in detail. Section IV gives the experimental results and analysis based on the miniSAR real dataset. Finally, Section V concludes this paper.

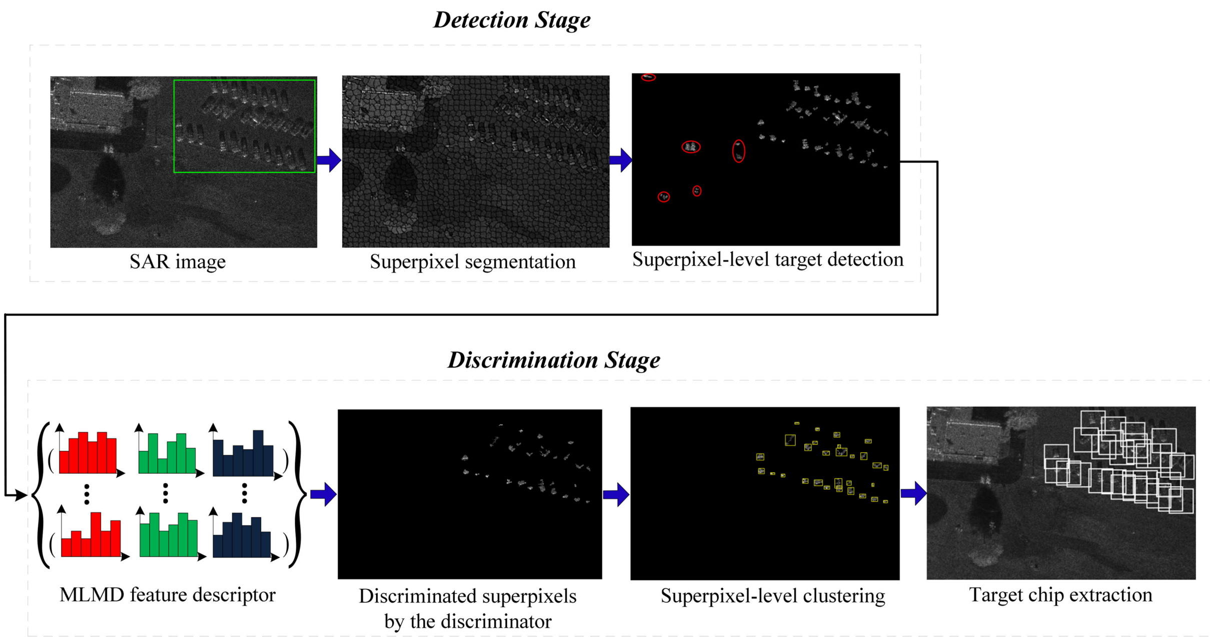

Fig. 4. Flowchart of the proposed method. Here, the detection stage is used to provide the candidate target superpixels for the discrimination stage, which can be seen as the preprocessing step for the superpixel-level SAR target discrimination.

II. Overview of the Proposed Method

Fig. 4 shows the flowchart of the proposed method, where we use a small SAR image as an example for the purpose of illustration. As shown in Fig. 4, we denote the target region using the green box in the original SAR image, and the other regions are the clutter, such as the buildings, trees, and grasslands. In the following, we describe each step of the proposed briefly.

1) *Superpixel segmentation:* The superpixel segmentation is a fundamental step for both the detection stage and the discrimination stage. Here, we employ the superpixel segmentation algorithm proposed by Yu *et al.* [6], where they introduced the generalized likelihood ratio dissimilarity measurement into simple linear iterative clustering [21] to make the algorithm more robust to the speckle noise in SAR images. In this example, the segmented superpixels are marked by the black curves.
2) *Superpixel-level target detection:* We employ a recently proposed superpixel-level target detection method [7], which is based on Bayesian-morphological saliency (BMS), to provide the candidate target superpixels for SAR target discrimination. In this example, most target superpixels are detected, and the clutter false alarm superpixels are marked by red ellipses. We hope to eliminate the clutter false alarm superpixels while retaining the target superpixels in the following discrimination stage.
3) *MLMD feature descriptor:* With the candidate target superpixels obtained by the target detection stage, we represent each superpixel via the MLMD feature descriptor, which can give a comprehensive description of the superpixel. This step is critical for the whole proposed target discrimination method, and the details about the MLMD feature descriptor will be further introduced in Section III.
4) *Discriminated superpixels by the discriminator:* The extracted MLMD features are input into the support vector machine (SVM) [27] discriminator to obtain the discriminated superpixels. In this example, we can see that all the clutter false alarm superpixels generated by the detection stage are excluded, and most target superpixels are retained.
5) *Superpixel-level clustering:* We employ the superpixel-level clustering method proposed in [6] to group the discriminated superpixels into several target clusters. In this example, each cluster is denoted by a yellow box.
6) *Target chip extraction:* With the target clusters obtained by the superpixel-level clustering, the target chips are extracted from the original SAR image based on the centroids of target cluster. It is worth noting that the traditional SAR discrimination methods first extract the chips and then implement target discrimination, while in our proposed SAR discrimination method, we first implement target discrimination at the superpixel-level, and then extract the chips, which can avoid the shortcomings of the traditional target discrimination methods at the chip-level. In this example, the extracted chips are denoted by white boxes. As shown in this example, almost all the targets are detected without any false alarms.

III. MLMD Feature Descriptor

A. Pixel-Level Multidomain Feature Description

Fig. 5 shows the diagram of the pixel-level multidomain feature description. As shown in Fig. 5, the input SAR image is analyzed through three domains, i.e., the texture domain, the multiscale amplitude domain, and the subaperture scattering domain. Each domain is described as follows.

1) Texture Domain: SIFT features are widely used in the optical image field. However, the SIFT features for optical images

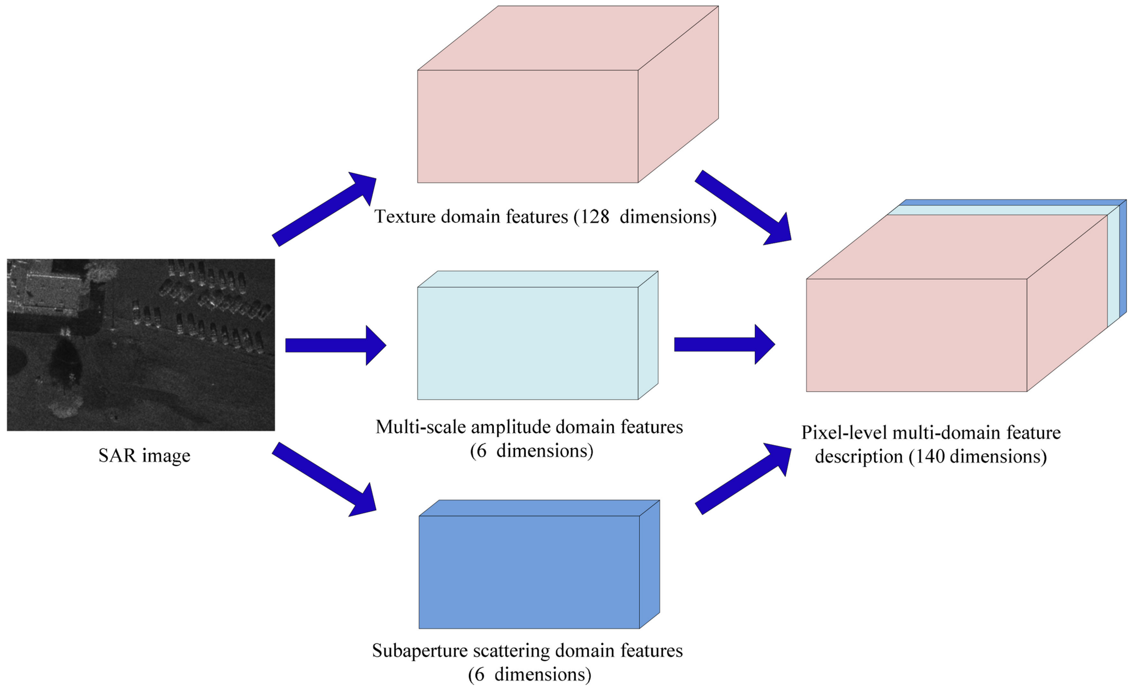

Fig. 5. Diagram of the pixel-level multidomain feature description.

are not robust to the speckle noise in SAR images. To make the SIFT features robust to the speckle noise, Dellinger *et al.* [24] proposed the SAR-SIFT features by defining a new way to calculate the gradient. In this paper, we employ the SAR-SIFT features that have 128 dimensions to obtain the texture domain features for each pixel.

2) Multiscale Amplitude Domain: The image Gaussian pyramid can reflect the amplitude characteristics of the objects with different sizes in different scales. Based on this fact, Wang *et al.* [8] proposed a SAR target detection method via the task-dependent scale selection from the amplitude Gaussian pyramid of the SAR image, where they mentioned that the targets of interest and the clutter usually have different sizes and present different amplitude characteristics in the amplitude Gaussian pyramid. In this paper, to obtain the multiscale amplitude domain features, we adopt the amplitude Gaussian pyramid with six scales and interpolate each image in the Gaussian pyramid into the same size as the input original SAR image.

3) Subaperture Scattering Domain: Subaperture images correspond to the observations under different azimuth look angles, and they can be obtained by subaperture decomposition technology [25]. The targets and the clutter present different scattering characteristics in different subaperture images [25], [26], i.e., the targets usually have high scattering intensity in all subaperture images, while the clutter does not have this characteristic. In this paper, we decompose the original complex-valued SAR image into six subaperture images.

B. Superpixel-Level Feature Description

Fig. 6 shows the superpixel-level feature extraction procedure based on the BoVW model. As shown in Fig. 6, the current superpixel marked by the red color contains many pixels that are represented by the 140-dimensional feature vectors obtained by the pixel-level feature description. The superpixel-level feature extraction procedure is described as follows. First, to obtain the codebook (set of code words), we employ the k-means clustering algorithm to cluster the pixel-level descriptors of the training images into k clusters and regard each cluster center as a code word of the codebook. In Fig. 6, k is set to 6, which is only an example for illustration. In our experiment in Section IV, we will give a detailed description how to choose k in practice. Then, we assign each pixel within the superpixel to the closest code word by calculating the Euclidean distance between the pixel and all the code words. Finally, we count the code words of the pixels within the superpixel to generate a histogram of code words, i.e., the superpixel-level features.

C. Object-Level Feature Description

The contextual information is very important for image classification, which can increase the separability of different classes in the feature space. For the superpixels, incorporating the contextual information can reflect the object-level information for classification. To make full use of the contextual information of the superpixels, Vargas *et al.* [23] proposed a contextual superpixel description method for the optical remote sensing image classification, where they used color and texture features as the pixel-level descriptors. For SAR images, the SAR-SIFT features can reflect the texture information. However, there are no color features in the SAR images. As mentioned previously, in our proposed SAR target discrimination method, we use three domain features, i.e., the texture feature, the multiscale amplitude feature, and the subaperture scattering feature, which are fit

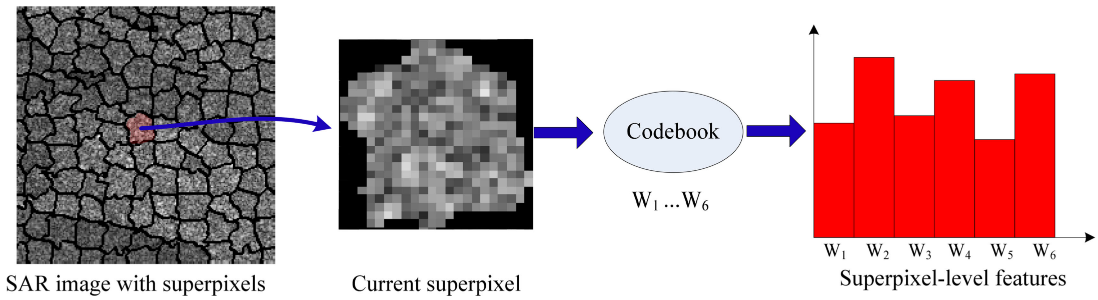

Fig. 6. Superpixel-level feature extraction procedure based on the BoVW model. Here, the superpixel with the red color in the SAR image is the current superpixel that is used to extract the superpixel-level features.

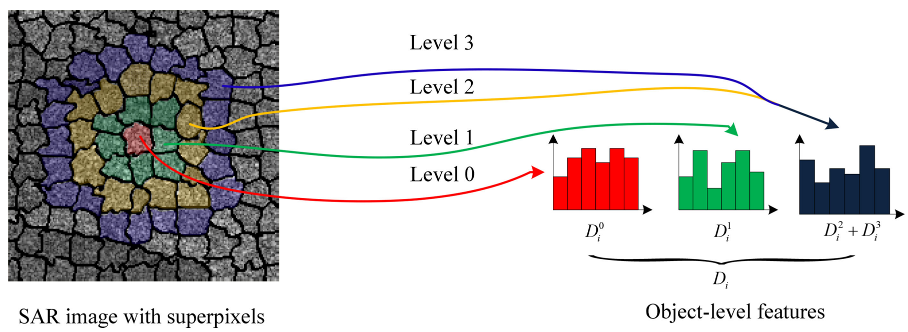

Fig. 7. Object-level feature extraction procedure that utilizes the contextual information from the nearby superpixels of the current superpixel.

for SAR images, as the pixel-level descriptors, which can give a comprehensive description of the SAR image pixels.

Fig. 7 shows the object-level feature extraction procedure that utilizes the contextual information from the nearby superpixels of the current superpixel S_i, where S_i is denoted by the red color. As shown in Fig. 7, we incorporate three levels (Level 1–Level 3) of the contextual information of S_i (Level 0). Here, Level 1–Level 3 are the 1–3 neighborhood superpixels of S_i, and they are denoted by the green, yellow, and blue colors, respectively, in the SAR image in Fig. 7. The object-level feature descriptor of S_i is defined as

$$D_i = \left(D_i^0, D_i^1, D_i^2 + D_i^3\right) \tag{1}$$

where D_i^0 is the superpixel-level feature descriptor of S_i, D_i^1 is the sum of superpixel-level feature descriptors of all the 1-neighborhood superpixels of S_i, D_i^2 is the sum of the superpixel-level feature descriptors of all the 2-neighborhood superpixels of S_i, and D_i^3 is the sum of the superpixel-level feature descriptors of all the 3-neighborhood superpixels of S_i. As shown in (1), D_i contains three components, where the position of each component denotes the spatial position information of the nearby superpixels of S_i. As in [23], we also aggregate the 2- and 3-neighborhood descriptors into one component of D_i, since the superpixels far away from the current superpixel are more likely to be the background and the position information of them is not as important as that of the superpixels close to the current superpixel.

IV. Experimental Results and Analysis

A. Experimental Data Description

The miniSAR real SAR dataset is acquired by Sandia National Laboratories, USA.[1] We select eight SAR images that have the vehicle targets from the miniSAR dataset for our experiments, where the resolutions of these SAR images are $0.1\,\text{m} \times 0.1\,\text{m}$. Here, four SAR images are selected as the training set and four SAR images are selected as the test set. Here, we only give the four SAR images for test and their corresponding manually labeled ground truths in Figs. 8 and 9, respectively. For convenience, we refer to the four SAR images as Images I–IV, respectively. As shown in Fig. 8, the scenes of these SAR images are very complex, where there are vehicles, helicopters, buildings, roads, grasslands, trees, and so on in these SAR images. Moreover, Table I shows the main SAR parameters of Images I–IV, including the working frequency band, spatial resolution, acquisition date and time, equivalent number looks (ENL), and image size.

B. Comparison Methods

Traditional SAR discrimination methods usually employ the probability of correct discrimination with respect to the same chip-level database for performance evaluation. However, the SAR discrimination database for the proposed discrimination method is at the superpixel-level; thus, it is hard to compare different discrimination methods directly. Moreover, it is

1.http://www.sandia.gov/radar/imagery

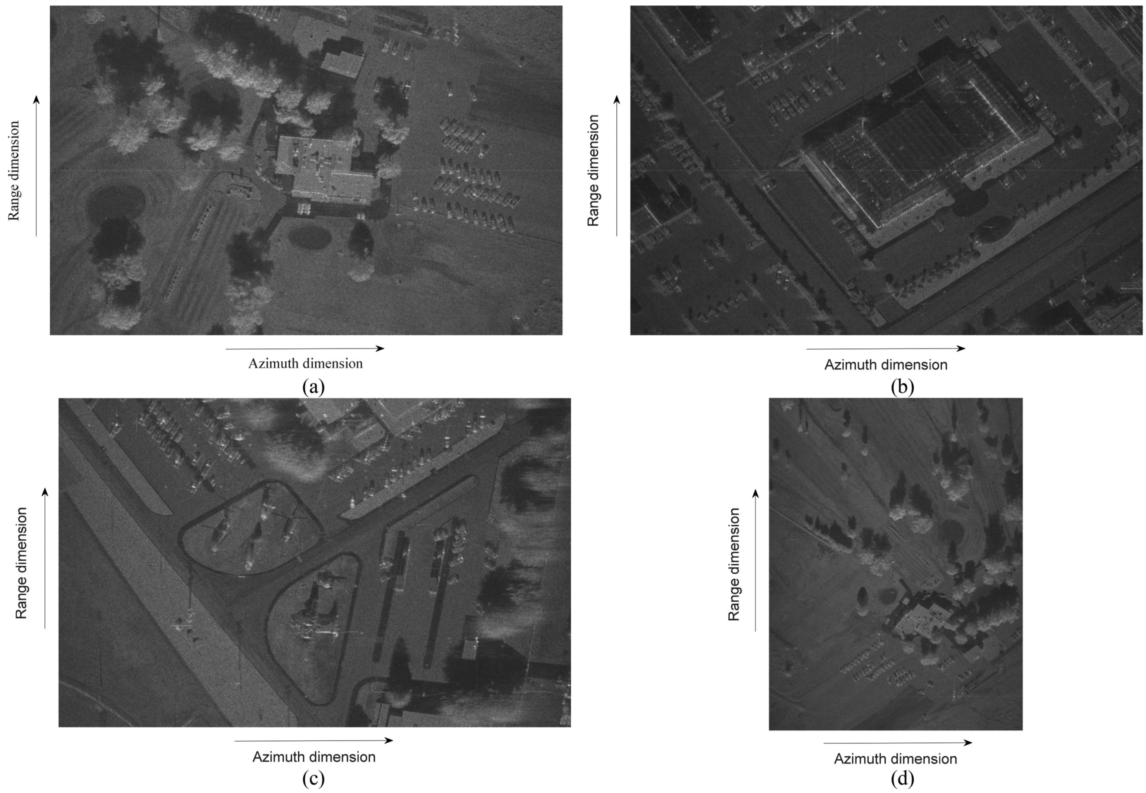

Fig. 8. Four SAR images from miniSAR dataset used for testing. (a) Image I. (b) Image II. (c) Image III. (d) Image IV.

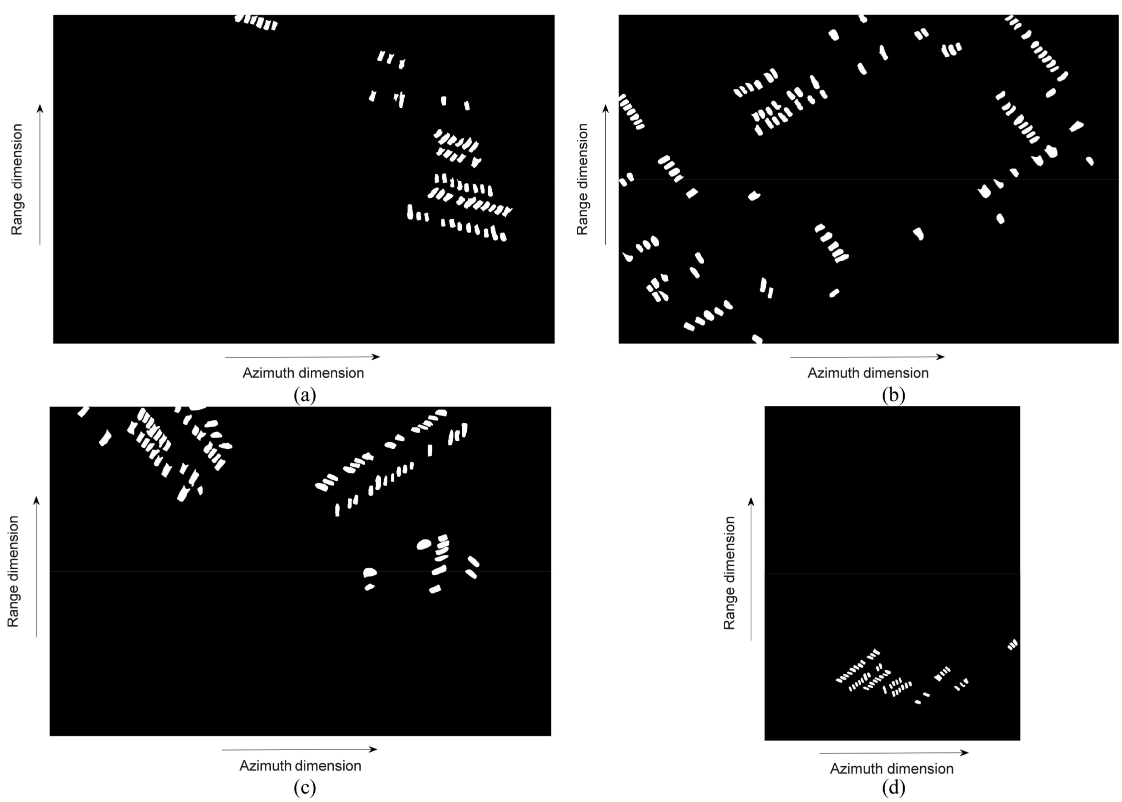

Fig. 9. Manually labeled ground truths of the four SAR images for testing. (a) Image I. (b) Image II. (c) Image III. (d) Image IV.

TABLE I
SAR ACQUISITION PARAMETERS OF FOUR TEST IMAGES

Image \ Parameters	Working frequency band	Spatial resolution	Acquisition date and time	ENL	Image size
Image I	Ku-Band	0.1m × 0.1m	16:13,5.31,2005	4.44	1638 × 2510
Image II			22:45,6.29,2005	3.37	
Image III			10:38,6.14,2005	4.40	
Image IV			11:27,6.29,2005	5.31	3274 × 2510

TABLE II
NUMBER OF DISCRIMINATION SAMPLES OBTAINED BY DIFFERENT METHODS

Sample acquisition methods	Training set		Test set	
	Number of the target samples	Number of the clutter samples	Number of the target samples	Number of the clutter samples
Chips obtained by CFAR	232	796	321	710
Chips obtained by BMS	317	581	374	508
Superpixels obtained by BMS	358	540	460	545

insufficient to evaluate different discrimination methods without considering the detection stage, since we are more interested in the whole performance after the detection and discrimination stages in the real SAR ATR system. To compare the different discrimination methods more fairly, we evaluate the final performance of the different methods after the detection and discrimination stages. In this experiment, we employ three methods as the comparison methods for the proposed method (BMS + MLMD features), and each comparison method is described as follows.

1) CFAR + CDFs: In the detection stage, we employ the classical two-parameter CFAR [1] as the detection method and then obtain the candidate target chips by the pixel-level clustering. As shown in Table II, there are 232 target chips and 796 clutter chips in the training set, and 321 target chips and 710 clutter chips in the test set. In the discrimination stage, we extract the CDFs for each candidate target chip and use the SVM as the discriminator.

2) BMS + CDFs: In the detection stage, we employ the BMS as the detection method and then obtain the candidate target chips by the superpixel-level clustering. As shown in Table II, there are 317 target chips and 581 clutter chips in the training set, and 374 target chips and 508 clutter chips in the test set. In the discrimination stage, we extract the CDFs for each candidate target chip and use the SVM as the discriminator.

3) BMS + MLTDFs: In the detection stage, we employ the BMS as the detection method and then obtain the candidate target superpixels for discrimination. As shown in Table II, there are 358 target superpixels and 540 clutter superpixels in the training set, and 460 target superpixels and 545 clutter superpixels in the test set. Note that the training set and the test set of the proposed method are the same as those of the BMS + MLTDFs. In the discrimination stage, we extract the multilevel texture domain features (MLTDFs) for each candidate target superpixels and use the SVM as the discriminator. Here, this method is the same as the proposed discrimination method except that the MLTDFs only use the texture domain information.

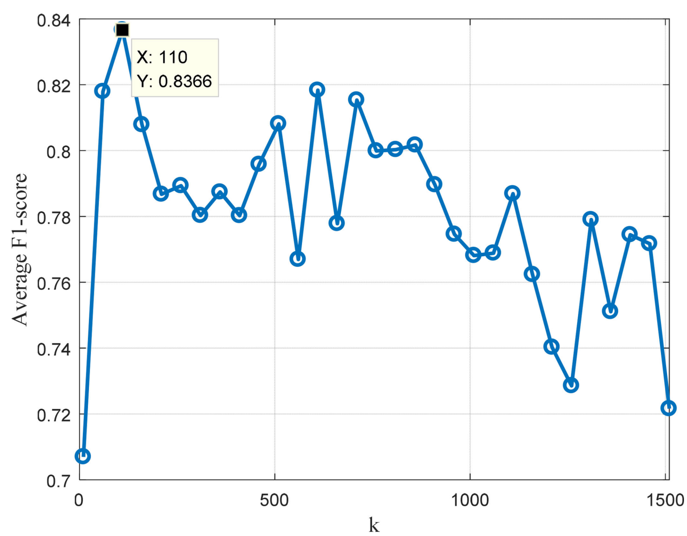

Fig. 10. Variation curve of the average $F1$ - score for four test SAR images with increasing k. Here, the maximum average $F1$ - score is 0.8366 when k is 110.

As shown in Table I, the number of the clutter samples is much more than that of the target samples in the three training sets. Here, to balance the numbers of the target samples and clutter samples in each training set, we select the same number of clutter samples as that of the target samples for training the SVM discriminators.

C. Evaluation Criteria

We employ $F1$ - score [4] to evaluate the performance of the different discrimination methods with target detection and discrimination. The definition of $F1$ - score is as follows:

$$F1 \text{ - score} = \frac{2 \times \text{precision} \times \text{recall}}{\text{precision} + \text{recall}} \quad (2)$$

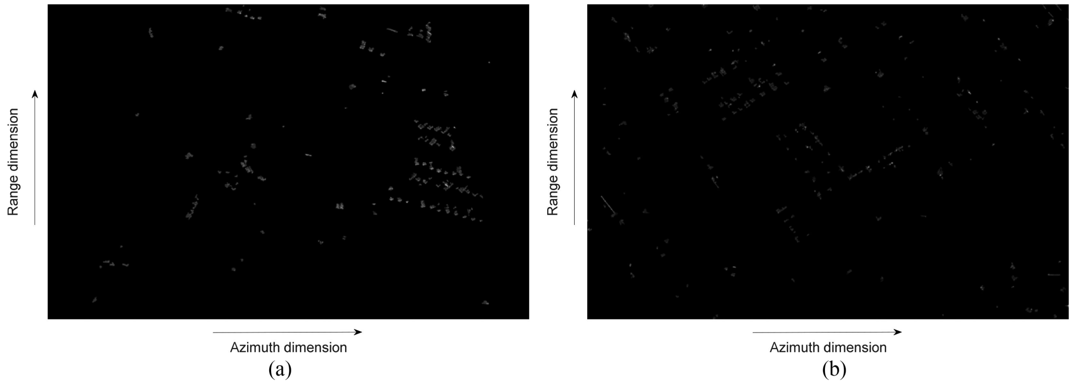

Fig. 11. Superpixel-level target detection results. (a) Image I. (b) Image II.

$$\text{recall} = \frac{\text{true positive}}{\text{total number of targets}} \tag{3}$$

$$\text{precision} = \frac{\text{true positive}}{\text{detected objects}}. \tag{4}$$

The larger the $F1$ - score, the better the performance. Here, the target is true positive only if the intersection area between the chip and the ground truth of this target is equal to or larger than 50% of the ground truth area.

D. Parameters Setting

1) Detection Stage: For the two-parameter CFAR, the false alarm rate p_{fa} is set to 0.004, and for the BMS detection method, the target confidence level α [7] is also set to 0.004.

2) Discrimination Stage: For the MLTDFs and the proposed MLMD features, the number of code words k for constructing the superpixel-level features is set to 110 (we will analyze the reasons why we choose this value in the following). For all the discrimination methods, the linear SVM is employed as the discriminator, where the penalty factor C is set to 5. Here, the dimension of the CDFs is 34, and the dimension of the MLMD features is 330.

3) Selection of k: In practice, if the dimension of the feature is much larger than that of the sample size, it will cause dimensionality and overfitting problems [28]. In our experiments, for the proposed method, the number of the training samples of each class for SVM is 358. Moreover, the dimension of the proposed MLMD features is $k \times 3$. Therefore, to avoid the dimensionality and overfitting problems, the feature dimension should be smaller than the samples of each class (i.e., $k \times 3$ should be smaller than 358); meanwhile, to avoid generating too much information loss, the feature dimension should not be too small.

In the following, we change k as [10:50:1510] to execute the parameter sensitivity analysis of k.

Fig. 10 shows the variation curve of the average $F1$ - score for four test SAR images with increasing k. As shown in Fig. 10, first the average $F1$ - score increases with k, then it reaches its maximum when k is 110, and finally it gradually decreases with k. Therefore, we set k to 110 in our experiments,

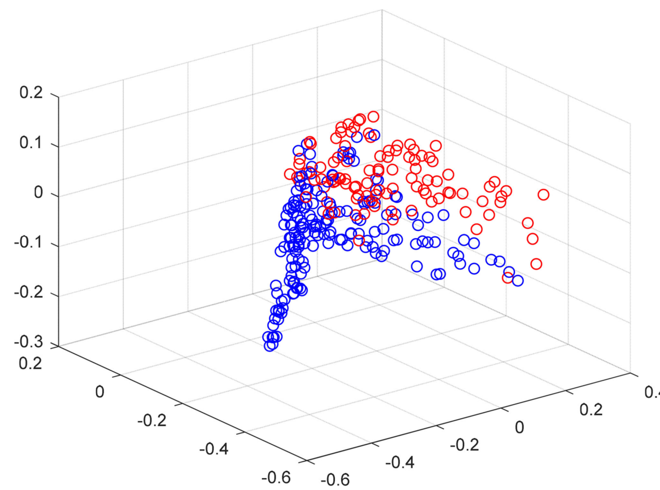

Fig. 12. Scatterplot of the MLMD feature descriptor. Here, the samples are from Images I and II, and each sample (330 dimensions) is projected into three dimensions by PCA. Moreover, the red points represent the targets, and the blue points represent the clutter.

which is also in accordance with our analysis presented above (i.e., the feature dimension is 110×3, which is smaller than the sample size of each class, i.e., 358, and not too small).

E. Performance of the Proposed Method in Different Steps

As shown in Fig. 4, there are five steps from the superpixel-level detection to the final target chip extraction, i.e., superpixel-level target detection, MLMD feature descriptor, discriminated superpixels by the discriminator, superpixel-level clustering, and target chip extraction. In the following, we take Images I and II as examples to see how the proposed method performs at the five steps.

Figs. 11–15 show the results of the five steps, respectively. As shown in Fig. 11, there are some false alarm superpixels by the superpixel-level target detection step. Then, the following steps of the discrimination stage are expected to eliminate these false alarms while retaining most true target superpixels.

As shown in Fig. 12, we project each MLMD feature descriptor into three dimensions by principal component analysis (PCA) for visualization, from which we can see that the target

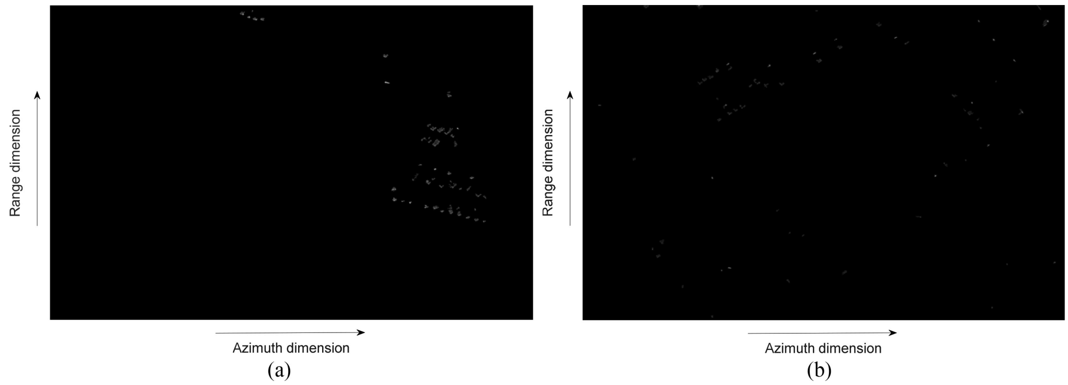

Fig. 13. Discriminated superpixels by the discriminator. (a) Image I. (b) Image II.

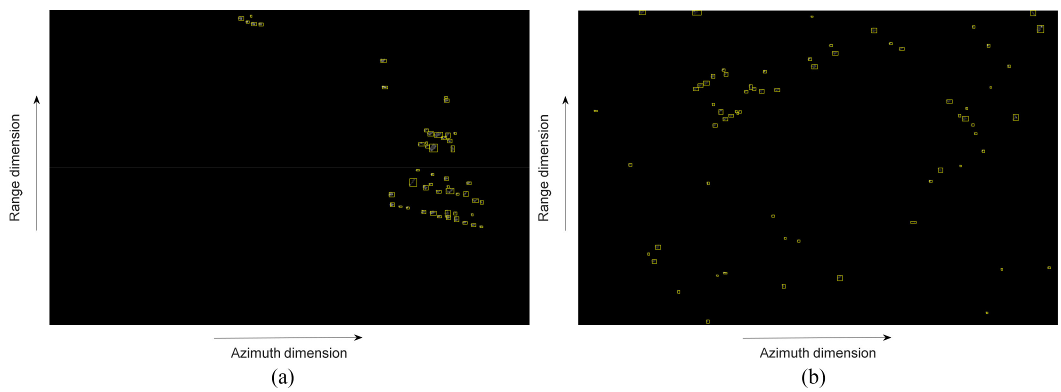

Fig. 14. Superpixel-level clustering results. (a) Image I. (b) Image II.

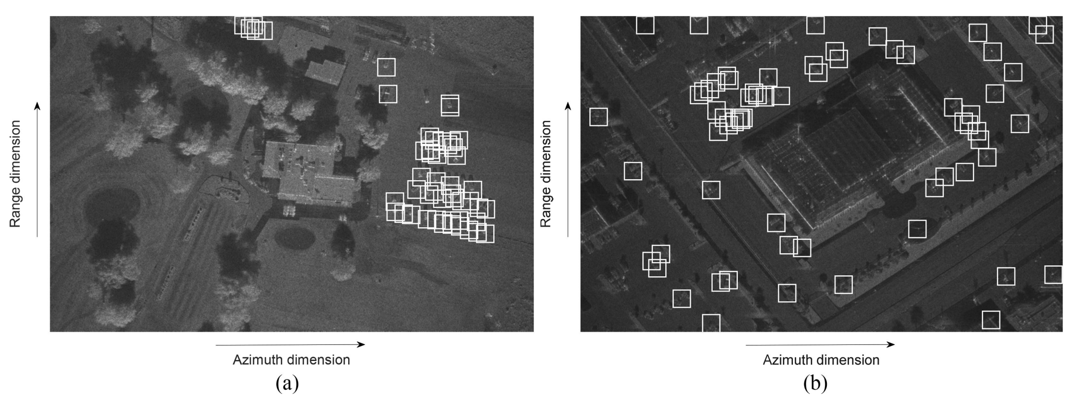

Fig. 15. Target chip extraction results. (a) Image I. (b) Image II.

and the clutter have relative good separability. Fig. 13 shows the discriminated superpixels by the discriminator. Comparing Figs. 13 and 11, we can see that the discriminator can eliminate a large number of false alarms and retain most target superpixels. As shown in Fig. 14, we merge the discriminated superpixels that belong to one target into a cluster by the superpixel-level clustering algorithm. Fig. 15 shows the target chip extraction results based on the clustering results in Fig. 14, from which we can see that most chips are the target and only a few false alarm chips occur.

To further illustrate the effectiveness of the proposed method, we also cluster the detected superpixels in Fig. 11 obtained by the detection stage and extract the chips based on the cluster center. Then, we quantitatively compare the chip-level results of the detection and discrimination stages. Table III shows the performance comparison of these two stages. As shown in Table III, the discrimination stage has much higher precision and $F1$ - score than the detection stage under the similar recall, which demonstrates the effectiveness of the discrimination stage of the proposed method.

TABLE III
PERFORMANCE COMPARISON OF DIFFERENT STAGES

Stage / Evaluation criteria	Detection stage		Discrimination stage	
	Image I	Image II	Image I	Image II
recall	1	0.7879	0.9074	0.6566
precision	0.4615	0.3805	1	0.8667
*F*1-*score*	0.6316	0.5132	0.9515	0.7471

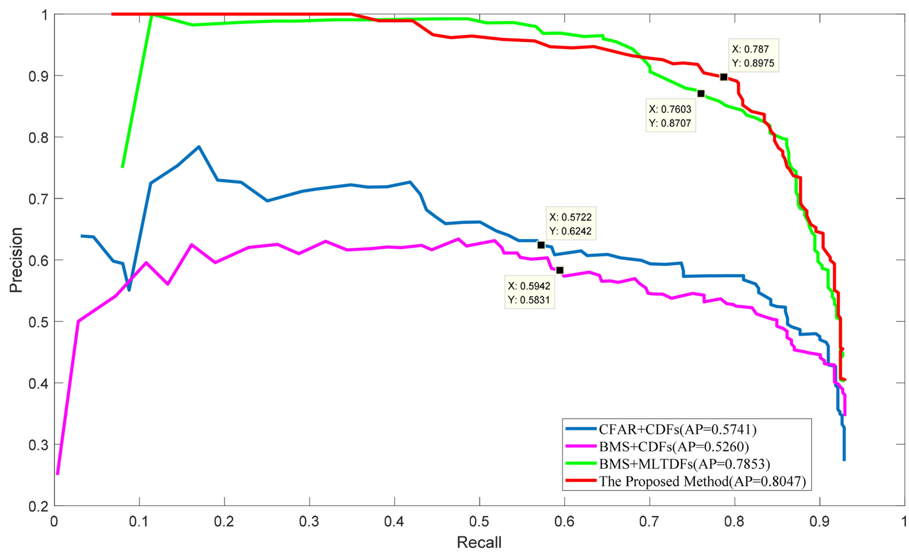

Fig. 16. PR curves of different methods. Here, the AP of CFAR + CDFs is 0.5741, the AP of BMS + CDFs is 0.5260, the AP of BMS + MLTDFs is 0.7853, and the AP of the proposed method is 0.8047. The data cursors on the curves denote the 281th operation points of the four methods respectively.

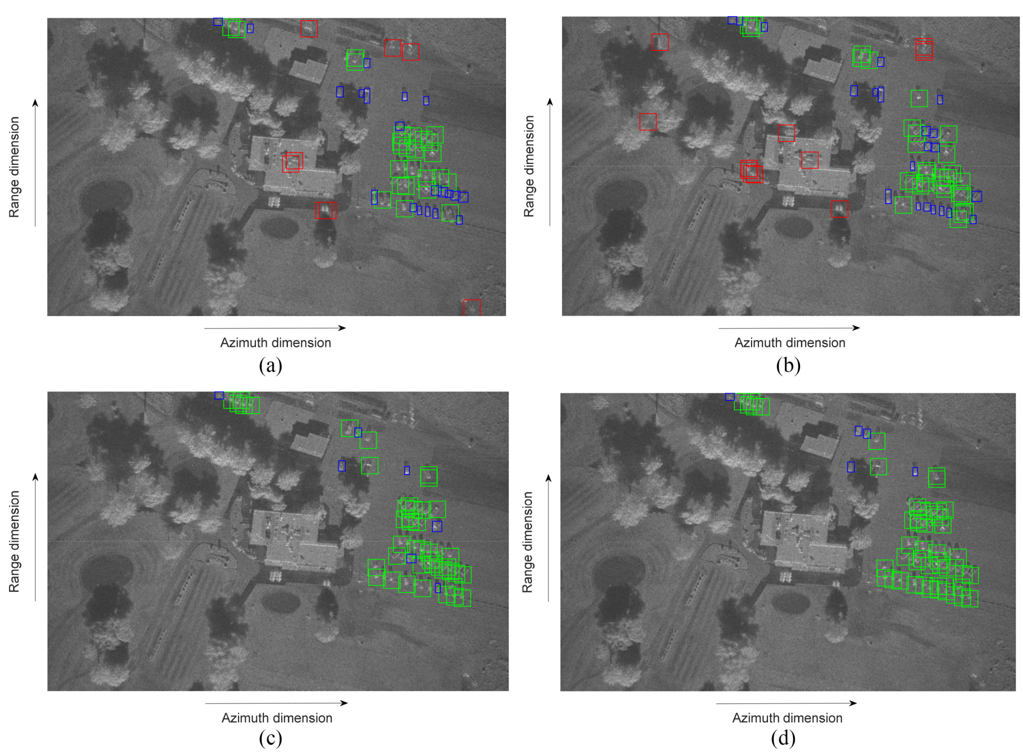

Fig. 17. Discrimination results of Image I using different methods. (a) CFAR + CDFs. (b) BMS + CDFs. (c) BMS + MLTDFs. (d) Proposed method. Here, the green square boxes denote the detected target chips, the red square boxes denote the clutter false alarm chips, and the blue rectangle boxes denote the missed targets.

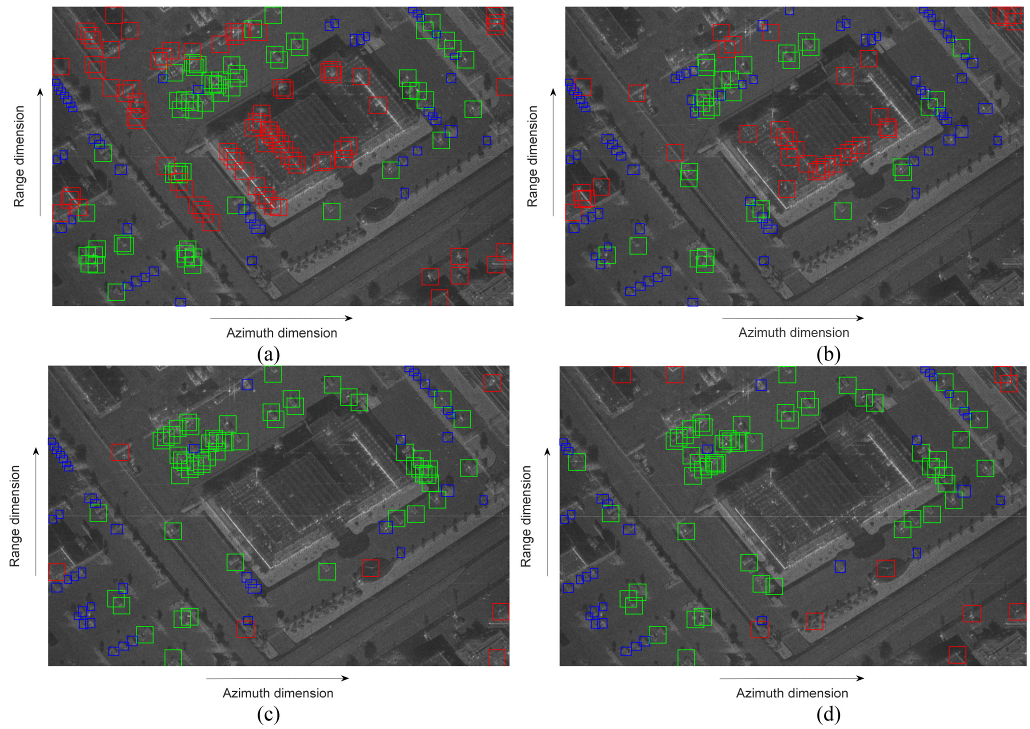

Fig. 18. Discrimination results of Image II using different methods. (a) CFAR + CDFs. (b) BMS + CDFs. (c) BMS + MLTDFs. (d) Proposed method. Here, the green square boxes denote the detected target chips, the red square boxes denote the clutter false alarm chips, and the blue rectangle boxes denote the missed targets.

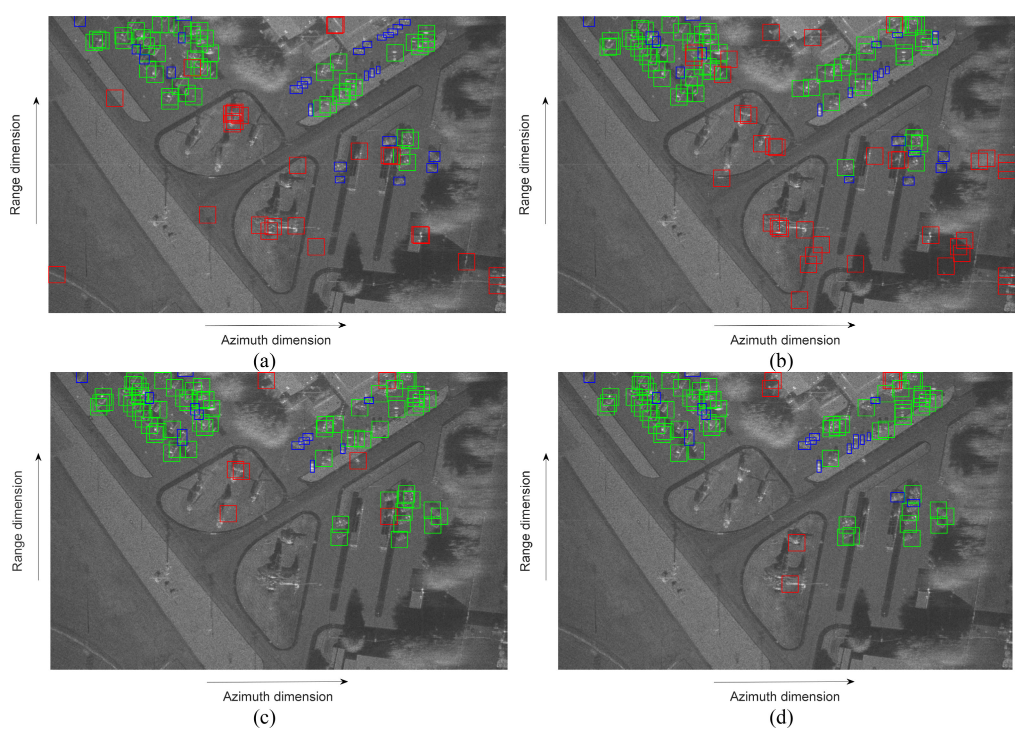

Fig. 19. Discrimination results of Image III using different methods. (a) CFAR + CDFs. (b) BMS + CDFs. (c) BMS + MLTDFs. (d) Proposed method. Here, the green square boxes denote the detected target chips, the red square boxes denote the clutter false alarm chips, and the blue rectangle boxes denote the missed targets.

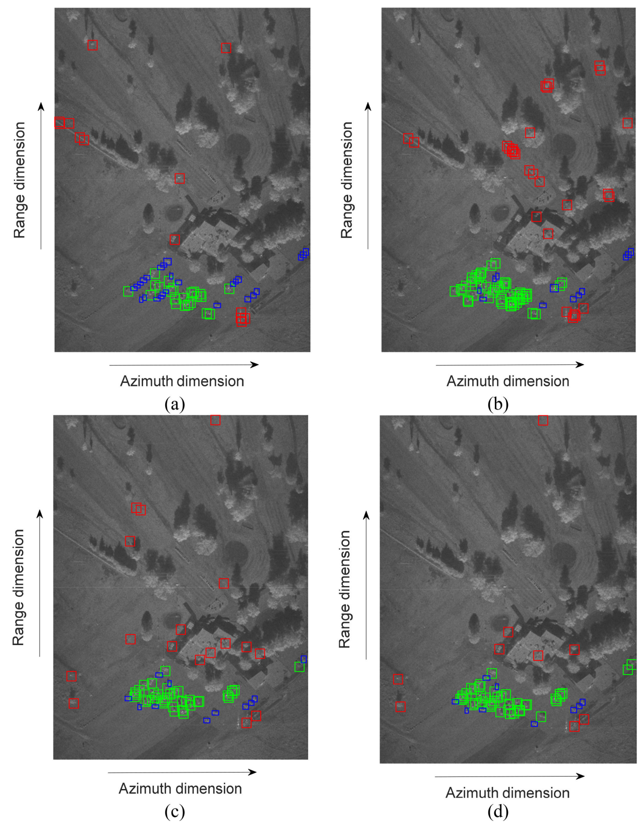

Fig. 20. Discrimination results of Image IV using different methods. (a) CFAR + CDFs. (b) BMS + CDFs. (c) BMS + MLTDFs. (d) Proposed method. Here, the green square boxes denote the detected target chips, the red square boxes denote the clutter false alarm chips, and the blue rectangle boxes denote the missed targets.

F. Comparison of Different Methods

1) PR Curves of Different Methods: We rank the output scores of the test samples obtained by the SVM discriminator in a descending order. Then, we can obtain different operation points based on the ranked output scores and plot the precision versus recall (PR) curves [29] of different methods. Fig. 16 shows the PR curves of different methods. The closer to the top right the PR curve, the better the performance. As shown in Fig. 16, the PR curves of the BMS + MLTDFs method and the proposed method are much better than those of the two traditional discrimination methods. i.e., CFAR + CDFs and BMS + CDFs. Moreover, we calculate the area under the PR curve of each method to obtain their average precision (AP) [29]. The larger the AP, the better the performance. As shown in Fig. 16, the proposed method has the largest AP among all methods.

2) Qualitative Comparison of Different Methods: Since the total number of the targets in the test images is 275, we select a operation point whose ranked number is slightly larger than 275, the 281th operation point, from the PR curves shown in Fig. 16 to compare different methods. Figs. 17–20 show the discrimination results of Images I–IV at their 281th operation points, where the green square boxes denote the detected target chips, the red square boxes denote the clutter false alarm chips, and the blue rectangle boxes denote the missed targets. As shown in Figs. 17–20, both the BMS + MLTDFs method and the proposed method, which are implemented at the superpixel-level, have much fewer missed targets and false alarms than CFAR + CDFs and BMS + MLTDFs methods, which are the traditional target discrimination methods at the chip-level. For CFAR + CDFs and BMS + MLTDFs methods, the missed targets are mainly from the multiple target areas, and the false alarms are mainly from the man-made clutter areas, while the BMS + MLTDFs method and the proposed method can detect most of the targets in the multiple target areas and almost generate no false alarms in the man-made clutter areas.

Fig. 21 shows the chip-level detection results of Image I by the detection stage of CFAR + CDFs. As shown in Fig. 21, the multiple target and partial target cases are generated simultaneously when we extract the square chips from the multiple target area in the SAR image. Thus, in the following, we will utilize several subimages including the multiple target areas as examples to further illustrate the effectiveness of the proposed method in dealing with the multiple/partial target cases.

3) Example of the Multiple/Partial Target Cases: Fig. 22 shows three subimages including multiple target areas, where multiple/partial target cases may easily be generated when we extract the square chips around the targets. In the following, we take these subimages as examples to further illustrate the effectiveness of the proposed method in dealing with the multiple/partial target cases.

Figs. 23 and 24 show the discrimination results of three subimages using CFAR + CDFs, BMS + CDFs, BMS + MLTDFs, and proposed methods. Note that CFAR + CDFs and BMS + CDFs methods are chip-level methods, while the BMS + MLTDF method and the proposed method are superpixel-level methods. As shown in Figs. 23 and 24, superpixel-level methods have much fewer missed targets than chip-level methods in the multiple target areas

4) Quantitative Comparison of Different Methods: Table IV shows quantitative comparison results of different methods at their 281th operation points. Note that the bold values in Table IV denote the largest F1-score for each SAR image or the largest average F1-score for the four SAR images. As shown in Table IV, both the BMS + MLTDFs method and the proposed method, which are implemented at the superpixel-level, have much higher recalls, precisions and $F1$ - scores than CFAR + CDFs and BMS + CDFs methods, which are the traditional target discrimination methods implemented at the chip-level. The larger the recall, the less the number of missed targets. Thus, the BMS + MLTDFs method and the proposed method, which have larger recalls, generate much fewer missed targets than the traditional target discrimination methods. Moreover, as shown in Figs. 17–20, for the traditional target discrimination methods, the missed targets are mainly from the multiple/partial target cases in the multiple target areas, while the proposed method and the BMS + MLTDFs method can avoid missing most of these targets in the multiple target areas, which demonstrate the effectiveness of the proposed method in dealing with the multiple/partial target cases.

In addition, as shown in Table IV, the proposed method has larger recall and $F1$ - score than the BMS + MLTDFs method

Fig. 21. Chip-level detection results of Image I by the detection stage of CFAR + CDFs. Here, the white square boxes denote the candidate target chips, and the yellow box denotes the multiple target area.

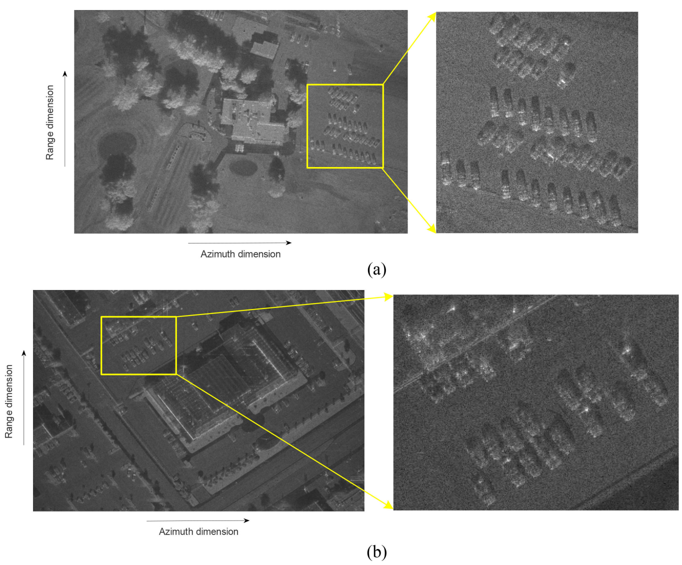

Fig. 22. Two subimage examples including multiple target areas. (a) Subimage of Image I. (b) Subimage of Image II.

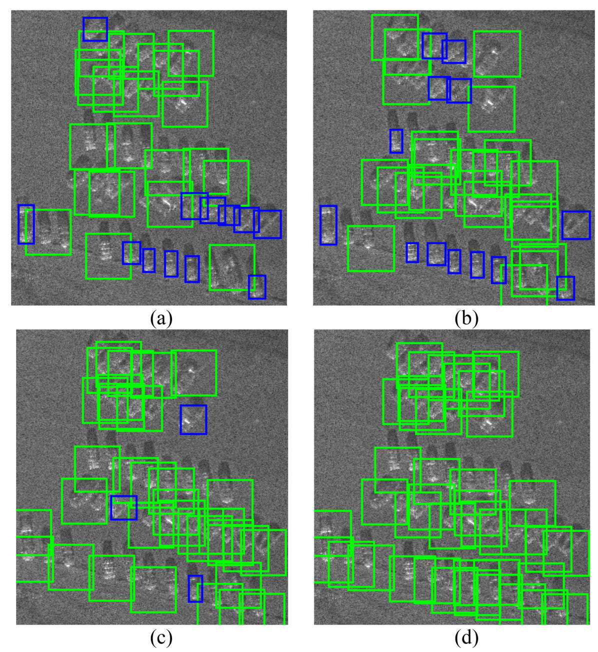

Fig. 23. Discrimination results of the subimage of Image I. (a) CFAR + CDFs. (b) BMS + CDFs. (c) BMS + MLTDFs. (d) The proposed method. Here the green square boxes denote the detected target chips, and the blue rectangle boxes denote the missed targets.

under the similar precision for Images I, II, and IV, and has a little smaller recall and $F1$ - score than the BMS + MLTDFs method under the similar precision for Image III. According to the discrimination results of the four test images, we can see that the proposed method has better discrimination performance than the BMS + MLTDFs method for three test images and a slightly worse discrimination performance than the BMS + MLTDFs method for one test image. Nevertheless, in terms of the average $F1$ - score of the four test images (Images I–IV), we can see from Table II that the proposed method has the highest average $F1$ - score among all the methods. In the following, we will analyze the reasons why the proposed method has a slightly worse performance than the BMS + MLTDFs method for Image III.

Compared with the proposed method, the BMS + MLTDFs method only considers the texture domain information of the targets, without considering the multiscale amplitude domain information. The multiscale amplitude domain information can reflect the size difference between targets and clutter via the intensity Gaussian pyramid of the original SAR image; however, it may lose its effectiveness when the target of interest is too close to the strong clutter in space. As shown in Fig. 25, Image III has some vehicle targets (in the small area denoted by the yellow box) that are very close to the strong ground clutter in space. Comparing Fig. 25(a) with (b), we can see that the proposed method has fewer false alarm chips and a little more missed targets than the BMS + MLTDFs method. As shown in Fig. 25(a), the missed targets of the proposed method are mainly from the vehicle targets in the small area denoted by the yellow box. The proposed method intends to utilize the size information reflected by the multiscale amplitude domain to discriminate the targets from the clutter; however, the vehicle targets in the yellow

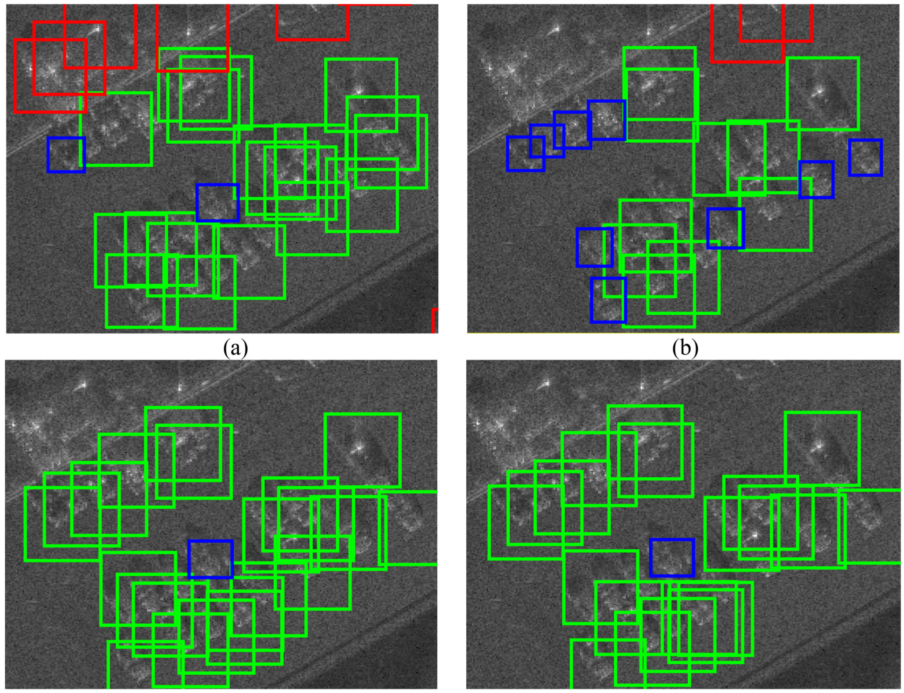

Fig. 24. Discrimination results of the subimage of Image II. (a) CFAR + CDFs. (b) BMS + CDFs. (c) BMS + MLTDFs. (d) Proposed method. Here, the green square boxes denote the detected target chips, the red square boxes denote the clutter false alarm chips, and the blue rectangle boxes denote the missed targets.

TABLE IV
PERFORMANCE COMPARISONS OF DIFFERENT METHODS

Methods	Evaluation criteria	Image I	Image II	Image III	Image IV
CFAR+CDFs	*recall*	0.6296	0.5354	0.6338	0.4902
	precision	0.8095	0.3955	0.6338	0.6579
	$F1$-*score*	0.7083	0.4549	0.6338	0.5618
	Average $F1$-*score*	0.5897			
BMS+CDFs	*recall*	0.6111	0.2828	0.7183	0.7647
	precision	0.7333	0.4308	0.5862	0.5821
	$F1$-*score*	0.6667	0.3415	0.6456	0.6610
	Average $F1$-*score*	0.5787			
BMS+MLTDFs	*recall*	0.8704	0.5556	0.8310	0.7843
	precision	1	0.8871	0.8939	0.7018
	$F1$-*score*	0.9307	0.6832	**0.8613**	0.7407
	Average $F1$-*score*	0.8040			
The proposed method	*recall*	0.9074	0.6566	0.7606	0.8235
	precision	1	0.8667	0.9000	0.8235
	$F1$-*score*	**0.9515**	**0.7471**	0.8244	**0.8235**
	Average $F1$-*score*	**0.8366**			

boxes and the strong ground clutter are almost being merged into an object with a larger size; thus, the proposed method may regard the targets denoted by the yellow boxes as the clutter and miss these targets. Nevertheless, this case is not common in the other test SAR images. When the targets are separated from the strong clutter in space, the proposed method can be effective. For example, Fig. 26(a) and (b) shows the discrimination results of Image II using the proposed method and BMS + MLTDFs method, respectively. As shown in Fig. 26(a), Image II has some vehicle targets (in the small area denoted by the yellow box) that

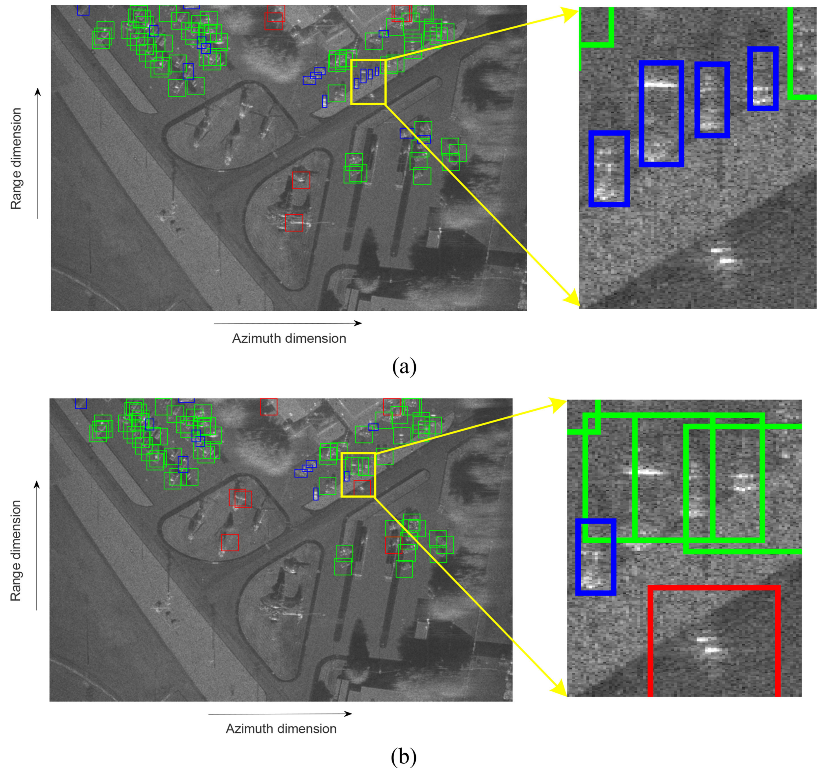

Fig. 25. Discrimination results of Image III using the proposed method and the BMS + MLTDFs method. (a) Proposed method. (b) BMS + MLTDFs method. Here, the green square boxes denote the detected target chips, the red square boxes denote the clutter false alarm chips, and the blue rectangle boxes denote the missed targets. Moreover, the yellow boxes denote the small area including the targets missed by the proposed method while detected by the BMS + MLTDFs method.

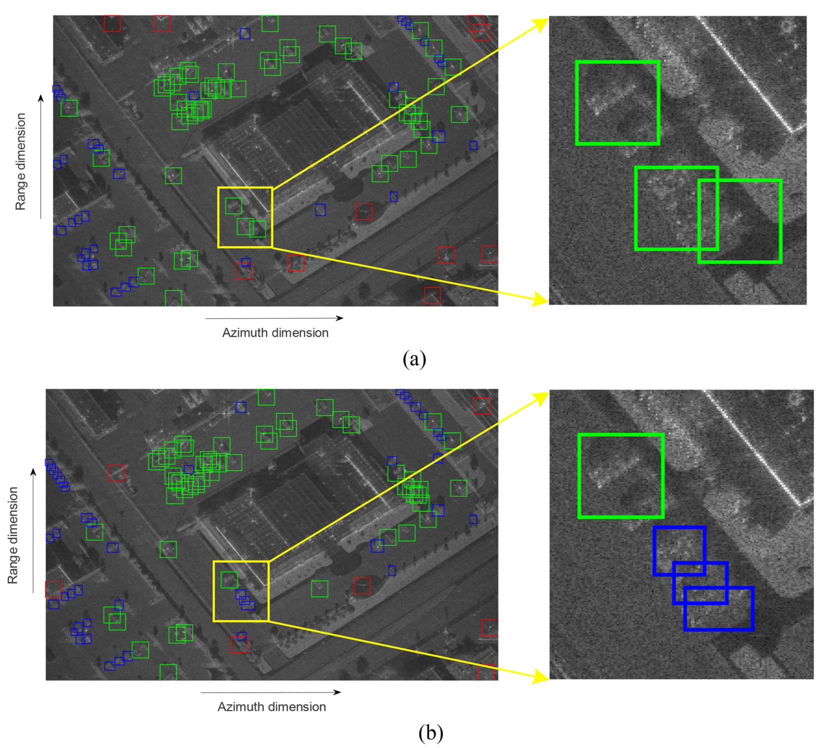

Fig. 26. Discrimination results of Image II using the proposed method and the BMS + MLTDFs method. (a) Proposed method. (b) BMS + MLTDFs method. Here, the green square boxes denote the detected target chips, the red square boxes denote the clutter false alarm chips, and the blue rectangle boxes denote the missed targets. Moreover, the yellow boxes denote the small area including the targets detected by the proposed method while missed by the BMS + MLTDFs method.

TABLE V
COMPUTATION LOAD OF DIFFERENT METHODS FOR IMAGE I

Method / Time (s)	CFAR+CDFs		BMS+CDFs		BMS+MLTDFs		The proposed method	
	Stage I	Stage II	Stage I	Stage II	Stage I	Stage II	Stage I	Stage II
Time of each stage	41	219	2808	160	2488	2338	2488	2601
Total time	260		2968		4826		5089	

are close to the strong ground clutter in space but are separated by some weak clutter. The proposed method can successfully detect these vehicle targets, while the BMS + MLTDFs method misses them.

The reasons for the superior discrimination performance of the proposed method are summarized as follows.

First, we use the superpixels rather than the chips as the elementary units for SAR target discrimination. Compared with the chips, the superpixels have more flexible sizes and shapes, which can adhere well to object boundaries, and each superpixel belongs to only one target or part of the background, not stretching over both of them. Thus, the proposed method can greatly reduce the occurrence of the multiple/partial target cases, which may easily occur for the SAR target discrimination methods using the chips as the elementary units. In addition, to extract the discrimination features for each superpixel, we utilize the synthetic information from the current superpixel and its nearby multiple superpixels, rather than only utilizing the information from the single chip as the CFAR + CDFs method and BMS + CDFs method did.

Second, we develop the MLMD feature descriptor to describe each superpixel discrimination sample. For both the BMS + MLTDFs method and the proposed method, the multilevel features based on the BoVW model are employed to describe each superpixel discrimination sample, which can give a finer feature description of each discrimination sample than the CDFs. Thus, both the BMS + MLTDFs method and the proposed method can exclude the building man-made clutter more easily than the traditional discrimination methods (the CFAR + CDFs method and BMS + CDFs method). In addition, as shown in Fig. 16 and Table IV, the performance of the proposed method is better than that of the BMS + MLTDFs method. This is due to that we extract the pixel-level descriptors in the multiple domains, i.e., the texture domain, the multiscale amplitude domain, and the subaperture scattering domain, instead of only extracting the pixel-level descriptors in the texture domain as the BMS + MLTDFs method did, which can give a more comprehensive description of the SAR images.

5) Computational Loads of Different Methods: We take one of the test images, Image I, as an example to compare the computational loads of different methods. We run the codes on a computer with 64-GB RAM and an Intel (R) Xeon (R) CPU E5-1630 v4 @3.70 Hz. The codes are written in MATLAB 2016b and not optimized. Table V shows the computational load of different methods for Image I, where Stage I denotes the detection stage and Stage II denotes the discrimination stage. As shown in Table V, the proposed method cost more time than the other methods. Although the proposed method is time-consuming under our current implementation, its speed can be accelerated via the parallel computing and code optimization.

V. CONCLUSION

This paper proposes a superpixel-level target discrimination method for high-resolution SAR images. In the proposed method, we employ the superpixels rather than the chips as the elementary units for SAR target discrimination. Moreover, we propose the MLMD feature descriptor to describe each superpixel. The experimental results using the challenging miniSAR real SAR dataset show the obvious performance improvement of the proposed discrimination method compared with the traditional discrimination methods. In the future work, we will explore more domains to further describe the superpixels, and test the proposed method with more real SAR images when we get other datasets.

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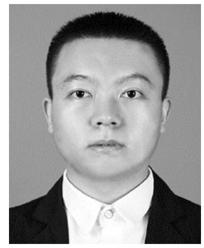

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